CS 4310 Operating Systems
Project #1 Simulating Job Scheduler and Performance Analysis

(Total: 100 points)

Student Name: _____

Date: _____

Important:
- Please read this document completely before you start coding.
- Also, please read the submission instructions (provided at the end of this document) carefully before submitting the project.

Project #1 Description:

Simulating Job Scheduler of the Operating Systems by programming the following four scheduling algorithms that we covered in the class:

a. First-Come-First-Serve (FCFS)
b. Shortest-Job-First (SJF)
c. Round-Robin with Time Slice = 2 (RR-2)
d. Round-Robin with Time Slice = 5 (PR-5)

You can use either Java or your choice of programming language for the implementation. The objective of this project is to help student understand how above four job scheduling algorithms operates by implementing the algorithms, and conducting a performance analysis of them based on the performance measure of their average turnaround times (of all jobs) for each scheduling algorithm using multiple inputs. Output the details of each algorithm's execution. You need to show which jobs are selected at what times as well as their starting and stopping burst values. You can choose your display format, for examples, you can display the results of each in *Schedule Table* or *Gantt Chart* format (as shown in the class notes). The project will be divided into three parts (phases) to help you to accomplish above tasks in in a systematic and scientific fashion: Design and Testing, Implementation, and Performance Analysis.

The program will read process burst times from a file (job.txt) – this file will be generated by you. Note that you need to generate multiple testing cases (with inputs of 5 jobs, 10 jobs and 15 jobs). A sample input file of five jobs is given as follows (burst time in ms):

[Begin of job.txt]
Job1
7
Job2
18
Job3
10
Job4
4
Job5
12
[End of job.txt]

Note: you can assume that

(1) There are no more than 30 jobs in the input file (job.txt).
(2) Processes arrive in the order they are read from the file for FCFS, RR-2 and RR-5.
(3) All jobs arrive at time 0.
(4) FCFS uses the order of the jobs, Job1, Job2, Job3, …

You can implement the algorithms in your choice of data structures based on the program language of your choice. Note that you always try your best to give the most efficient program for each problem. The size of the input will be limited to be within 30 jobs.

Submission Instructions:

- ***turn in the following @Canvas after the completion of all three parts, part 1, part 2 and part 3***
  (1) four program files (your choice of programming language with proper documentation)
  (2) this document (complete all the answers)

Part1
Design & Testing (30 points)

a. Design the program by providing pseudocode or flowchart for each CPU scheduling algorithm.

<Insert answers here>

b. Design the program correctness testing cases. Give at least 3 testing cases to test your program, and give the expected correct **average turnaround time** (for each testing case) in order to test the correctness of each algorithm.

<Complete the following table>

Testing case#	Input (table of jobs with its job# and length	Expected output for FCFS (√ if Correct after testing in Part 2)	Expected output for SJF (√ if Correct after testing in Part 2)	Expected output for RR-2 (√ if Correct after testing in Part 2)	Expected output for RR-5 (√ if Correct after testing in Part 2)
1 (5 jobs)	<insert answers here>	<insert answers here>	<insert answers here>	<insert answers here>	<insert answers here>
2 (10 jobs)	<insert answers here>	<insert answers here>	<insert answers here>	<insert answers here>	<insert answers here>
3 (15 jobs)	<insert answers here>	<insert answers here>	<insert answers here>	<insert answers here>	<insert answers here>

c. Design testing strategy for the programs. Discuss about how to generate and structure the randomly generated inputs for experimental study later in Part 3.

Hint 1: To study the performance evaluation of the four job scheduling algorithms, this project will use three different input sizes, 5 jobs, 10 jobs and 15 jobs. It is the easiest to use a random number generator for generating the inputs. Note that you need to decide the maximum value of job length (use at least 20). However, student should store each data set in various sizes and use the same data set for each job scheduling algorithm.

The performance of average Turnaround Time of each input data size (5 jobs, 10 jobs and 15 jobs) can be calculated after an experiment is conducted in 20 trail (with 20 input sets of jobs). We can denote the results as the set X which contains the 20 computed Turnaround Times of 20 trails, where X = {x_1, x_2, x_3 … x_{20}}, from the simulator.

For each data size (5 jobs, 10 jobs and 15 jobs):

$$\textit{Average Turnaround Time} = \frac{\sum_{i=1}^{20} Xi}{20}$$

The student should decide the maximum value of the job length (at least 20).

<Insert answers here>

Part 2
Implementation (30 points)

a. Code each program based on the design (pseudocode or flow chart) in Part 1(a).

<Generate four programs and stored them in four files, needed to be submitted>

b. Document the program appropriately.

<Generate documentation inside the four program files>

c. Test you program using the designed testing input data given in the table in Part 1(b), Make sure each program generates the correct answer by marking a "√" if it is correct for each testing case for each program column in the table. Repeat the process of debugging if necessary.

<Complete the four columns of the four algorithms in the table @Part 1(b)>

d. For each program, capture a screen shot of the execution (Compile&Run) using the testing case in Part 1(b) to show how this program works properly

<Insert totally four screen shots, one for each program, here>

By now, four working programs are created and ready for experimental study in the next part, Part 3.

Part 3
Performance Analysis (40 points)

a. Run each program with the designed randomly generated input data given in Part 1(c). Generate a table for all the experimental results for performance analysis as follows.

Input Size n jobs	Average of average turnaround times (FCFS Program)	Average of average turnaround times (SFJ Program)	Average of average turnaround times (RR-2)	Average of average turnaround times (RR-5)
5 jobs	*its average with 20 trials*	*its average with 20 trials*	*its average with 20 trials*	*its average with 20 trials*
10 jobs	*its average with 20 trials*	*its average with 20 trials*	*its average with 20 trials*	*its average with 20 trials*
15 jobs	*its average with 20 trials*	*its average with 20 trials*	*its average with 20 trials*	*its average with 20 trials*

b. Plot a graph of each algorithm, average turnaround time vs input size (# of jobs), and summarize the performance of each algorithm based on its own graph.

<Insert totally four graphs, one for each program, here>

<Write a summary>

Plot all four graphs on the same graph and compare the performance of all four algorithms. Rank four scheduling algorithms. Try giving the reasons for the findings.

<Insert four-graphs-in-one graph here>

<Write about explaining the results>

c. Conclude your report with the strength and constraints of your work. At least 100 words. (Note: It is reflection of this project. If you have a change to re-do this project again, what you like to keep and what you like to do differently in order get a better quality of results.)

<Write a conclusion about strength and constraints of your work here.>